

5.4 Electrochemistry Calculations & Applications (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.4 Electrochemistry Calculations & Applications (A Level Only)
Difficulty	Easy

Time allowed: 50

Score: /35

Percentage: /100

Question 1a

Table 1.1 shows the standard electrode potentials for the $\text{Mg}^{2+} / \text{Mg}$ and I_2 / I^- half-cells.

Table 1.1

Electrode reaction	E^θ / V
$\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$	+ 0.54
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	– 2.38

Calculate the standard cell potential for the electrochemical cell using the $\text{Mg}^{2+} / \text{Mg}$ and I_2 / I^- half-cells.

$$E^\theta = \dots\dots\dots \text{V}$$

[1 mark]

Question 1b

Write the equation for the reaction that occurs in the electrochemical cell described in part (a).

[1 mark]

Question 1c

Explain why the I_2 / I^- half cell is the positive pole for the electrochemical cell described in part (a).

[1 mark]

Question 2a

The electrochemical series was established by measuring the potential of various electrodes against the standard hydrogen electrode.

The electrochemical series can be used to identify oxidising and reducing agents.

In terms of electrons, explain what it meant by the terms oxidising agent and reducing agent.

[2 marks]

Question 2b

Table 2.1 shows a section of the electrochemical series.

Table 2.1

Electrode reaction	E^θ / V
$\text{Ba}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ba}$	-2.90
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	$+0.34$
$\text{F}_2 + 2\text{e}^- \rightleftharpoons 2\text{F}^-$	$+2.87$
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	$+1.52$
$\text{O}_2 + \text{H}_2\text{O} + 2\text{e}^- \rightleftharpoons \text{HO}_2^- + \text{OH}^-$	-0.08
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.76

Using Table 2.1, identify the strongest and weakest reducing agents.

[4 marks]

Question 2c

Using Table 2.1, identify the strongest and weakest oxidising agents.

[2 marks]

Question 3a

An electrochemical cell is set up to measure the electrode potential, E , for the $\text{Cu}^{2+} / \text{Cu}$ half-cell with a standard hydrogen electrode.

The standard electrode potential for the $\text{Cu}^{2+} / \text{Cu}$ half-cell is



State **two** factors that can affect the equilibrium position of the electrode reaction.

[1 mark]

Question 3b

A $0.750 \text{ mol dm}^{-3}$ solution of copper(II) chloride to make the $\text{Cu}^{2+} / \text{Cu}$ half-cell.

State the effects that this solution of copper(II) chloride has on the equilibrium position and electrode potential, E .

Equilibrium position

Electrode potential

[1 mark]

Question 3c

Identify the oxidised and reduced species for the $\text{Cu}^{2+} / \text{Cu}$ half-cell.

Oxidised species

Reduced species

[1 mark]

Question 3d

The Nernst equation is:

$$E = E^{\theta} + \frac{0.059}{z} \log_{10} \frac{[\text{oxidised species}]}{[\text{reduced species}]}$$

i)

To calculate the electrode potential at 298 K of the $\text{Cu}^{2+} / \text{Cu}$ half-cell, the Nernst equation can be revised to:

$$E = E^{\theta} + \frac{0.059}{z} \log_{10} [\text{oxidised species}]$$

Using your answer to part (c), suggest why the concentration of the reduced species does not feature in the revised Nernst equation calculation for the $\text{Cu}^{2+} / \text{Cu}$ half-cell.

[1]

ii)

Use the information in parts (a) and (b) and the revised Nernst equation to calculate the electrode potential at 298 K of the $\text{Cu}^{2+} / \text{Cu}$ half-cell.

$E = \dots\dots\dots \text{ V}$

[2]

[1 mark]

Question 4a

The standard free energy change can be calculated using the standard electrode potentials of the half-cells in an electrochemical cell.

$$\Delta G^{\ominus} = - n \times E^{\ominus}_{\text{cell}} \times F$$

Complete Table 4.1 to show what each term in the equation represents.

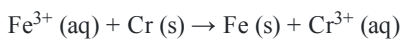
Table 4.1

Term	
$\Delta G^{\ominus}$	Standard free energy change
n	
$E^{\ominus}_{\text{cell}}$	
F	

[3 marks]

Question 4b

The following electrochemical cell is set up.



Use Table 4.2 to determine the **two** half-cell electrode reactions and their E^θ values.

Table 4.2

Electrode reaction	E^θ / V
$\text{Cr}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cr}$	- 0.91
$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	- 0.74
$\text{Cr}^{3+} + \text{e}^- \rightleftharpoons \text{Cr}^{2+}$	- 0.41
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	- 0.44
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	- 0.04
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	+ 0.77

[2 marks]

Question 4c

Use the information in part (a) and your answer to part (b), to calculate the standard Gibbs free energy change, in kJ mol^{-1} , for the electrochemical cell.

$$F = 96\,500 \text{ C mol}^{-1}$$

$$\Delta G^\theta = \dots\dots\dots \text{kJ mol}^{-1}$$

[4 marks]

Question 4d

Use your answer to part (c) to suggest whether the reaction is feasible. Explain your answer.

[1 mark]

Question 5a

Faraday's constant can be used to calculate the mass of a substance deposited at an electrode or the volume of gas liberated at an electrode during electrolysis.

A current of 1.65 A flows through molten lead(II) chloride for 10 minutes.

Write the equation for the electrolysis of lead(II) chloride, PbCl_2 .

[1 mark]

Question 5b

i)

Write the half-equation for the reduction of the lead(II) ion, Pb^{2+} .

[1]

ii)

The equation to calculate the charge transferred is $Q = I \times t$. The Faraday constant, $F = 96\,500 \text{ C mol}^{-1}$.

Use the information in part (a) to calculate:

The number of coulombs required to deposit one mole of lead at the cathode = C

The charge transferred during the electrolysis of the molten lead(II) bromide = C

The mass of lead deposited at the cathode = (g)

[5]

[6 marks]

Question 5c

i)

Write the half-equation for the oxidation of the chloride ion, Cl^- to form chlorine, Cl_2 .

[1]

ii)

The equation to calculate the charge transferred is $Q = I \times t$. The Faraday constant, $F = 96\,500 \text{ C mol}^{-1}$.

Use the information in part (a) and your answers to part (b) to calculate:

The number of coulombs required to form one mole of chlorine at the anode = C

The volume of chlorine released at the anode = (dm^3)

[2]

[3 marks]

Model Answers: Easy

1a

a) The standard cell potential for the electrochemical cell using the $\text{Mg}^{2+} / \text{Mg}$ and I_2 / I^- half-cells is:

- $E^\theta = 2.92 \text{ (V)}$; [1 mark]

[Total: 1 mark]

- **Remember:** $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}}$
- The I_2 / I^- half-cell is the reduction reaction as it has the highest standard electrode potential value
 - Therefore, $E^\theta_{\text{cell}} = 0.54 - (-2.38) = + 2.92\text{V}$
- For any feasible electrochemical cell, the standard cell potential is essentially the difference between the two standard electrode potential values

1b

b) The equation for the reaction that occurs in the electrochemical cell described in part (a) is:

- $\text{Mg} + \text{I}_2 \rightarrow \text{Mg}^{2+} + 2\text{I}^-$; [1 mark]

[Total: 1 mark]

- The $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$ electrode reaction / equation has the higher / more positive standard electrode potential value, which means that it is the reduction reaction
 - $\text{I}_2 + 2\text{e}^- \rightleftharpoons 2\text{I}^-$
- The $\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$ electrode reaction / equation has to be reversed so that it is the oxidation reaction
 - $\text{Mg} \rightleftharpoons \text{Mg}^{2+} + 2\text{e}^-$
- There is no need to multiply either equation by a coefficient because both equations involve two electrons
- The two electrode reactions / equations are then combined
 - $\text{I}_2 + 2\text{e}^- + \text{Mg} \rightleftharpoons 2\text{I}^- + \text{Mg}^{2+} + 2\text{e}^-$

- The electrons cancel out to give the overall equation for the feasible reaction
 - $\text{I}_2 + \text{Mg} \rightleftharpoons 2\text{I}^- + \text{Mg}^{2+}$

1c

c) The I_2 / I^- half cell is the positive pole for the electrochemical cell described in part (a) because:

- The I_2 / I^- half cell / it has the most positive standard electrode potential / E^θ value
OR

The I_2 / I^- half cell / it has a more positive standard electrode potential / E^θ value than the $\text{Mg}^{2+} / \text{Mg}$ half-cell; [1 mark]

[Total: 1 mark]

- The half-cell that is the positive pole is always the half-cell that has the highest / most positive standard electrode potential value

2a

a) The terms mean:

- Oxidising agent = a substance that causes another atom or ion to lose electrons
OR
a substance that accepts electrons; [1 mark]
- Reducing agent = a substance that causes another atom or ion to gain electrons
OR
a substance that donates electrons; [1 mark]

[Total: 2 marks]

- You need to know what oxidising and reducing agents are
- Examiners often comment about students have the two definitions mixed up

2b

b) The strongest and weakest reducing agents are:

- The strongest reducing agent = Ba / barium; [1 mark]

- The weakest reducing agent = F^- / fluoride; [1 mark]

[Total: 2 marks]

- The strongest reducing agent is the product from the electrode reaction with the most negative standard electrode potential
 - In this case, the most negative value is -2.90 which means that Ba is the strongest reducing agent
- The weakest reducing agent is the product from the electrode reaction with the most positive standard electrode potential
 - In this case, the most positive value is $+2.87$ which means that F^- is the weakest reducing agent

2c

c) The strongest and weakest oxidising agents are:

- The strongest oxidising agent = F_2 / fluorine; [1 mark]
- The weakest oxidising agent = Ba^{2+} / barium ions; [1 mark]

[Total: 2 marks]

- To determine the strongest and weakest reducing / oxidising agents, it can help to arrange the electrode reactions from the most positive to the most negative standard electrode potential:

Electrode reaction	$E^\ominus / V$
$F_2 + 2e^- \rightleftharpoons 2F^-$	$+2.87$
$MnO_4^- + 8H^+ + 5e^- \rightleftharpoons Mn^{2+} + 4H_2O$	$+1.52$
$Cu^{2+} + 2e^- \rightleftharpoons Cu$	$+0.34$
$O_2 + H_2O + 2e^- \rightleftharpoons HO_2^- + OH^-$	-0.08
$Zn^{2+} + 2e^- \rightleftharpoons Zn$	-0.76
$Ba^{2+} + 2e^- \rightleftharpoons Ba$	-2.90

- The strongest oxidising agent is the reactant from the electrode reaction with the most positive standard electrode potential
 - In this case, the most positive value is $+2.87$ which means that F_2 is the strongest oxidising agent
- The weakest oxidising agent is the reactant from the electrode reaction with the most

negative standard electrode potential

- In this case, the most negative value is -2.90 which means that Ba^{2+} is the weakest oxidising agent

3a

a) **Two** factors that can affect the equilibrium position of the electrode reaction are:

- Temperature

AND

Concentration; [1 mark]

[Total: 1 mark]

- You should know that temperature and concentration affect the equilibrium position for half-cell reactions
- You also need to be able to apply Le Chatelier's principle to half-cell equations

3b

b) The effects that this solution of copper(II) chloride has on the equilibrium position and electrode potential, E , are:

- Equilibrium position = moves to the left / reactants; [1 mark]
- Electrode potential = become less positive; [1 mark]

[Total: 2 marks]

- When a standard electrode potential is positive, the equilibrium position sits to the right-hand side
 - The higher the standard electrode potential, the further to the right the equilibrium position is
- Using this idea:
 - If the equilibrium position shifts to the left-hand side, the standard electrode potential value decreases
 - Similarly, if the standard electrode potential value decreases, then the equilibrium position shifts to the left-hand side
- Standard conditions for electrochemistry require a 1.00 mol dm^{-3} ion concentration
 - The Cu^{2+} ion concentration given in the question is $0.750 \text{ mol dm}^{-3}$, which

means that the concentration of Cu^{2+} ions has decreased

- Applying Le Chatelier's principle, a decrease in the concentration of a reactant results in the equilibrium shifting to the left-hand side
- This equilibrium shift to the left-hand side causes a decrease in the electrode potential, E

3c

c) The oxidised and reduced species for the $\text{Cu}^{2+} / \text{Cu}$ half-cell are:

- Oxidised species = Cu^{2+}

AND

Reduced species = Cu ; [1 mark]

[Total: 1 mark]

- The oxidised species is Cu^{2+} as it has a higher oxidation number (+2)
- The reduced species is Cu as it has a lower oxidation number (0)
- It is important to be able to correctly identify the oxidised and reduced species so that you put their values in the correct position in the Nernst equation

3d

d)

i) The concentration of the reduced species does not feature in the revised Nernst equation because:

- The reduced species is not in solution

AND

Therefore, it does not have a concentration; [1 mark]

ii) To calculate the electrode potential at 298K of the $\text{Cu}^{2+} / \text{Cu}$ half-cell:

- $E = + 0.34 + \frac{0.059}{2} \log_{10} [0.0750]$; [1 mark]
- $E = 0.31 \text{ (V)}$; [1 mark]

[Total: 3 marks]

- The specification states that you have to be able to use the $E = E^\theta + \frac{0.059}{z} \log_{10} \frac{[\text{oxidised species}]}{[\text{reduced species}]}$ version of the Nernst equation
 - The specification is not clear on whether you are expected to remember this equation or not, although one of the course textbooks states that you may need to be able to recall this equation
- If the oxidised species or reduced species does not have a concentration then it is replaced by the number 1

4a

a) The completed table is:

Term	
ΔG^θ	Standard free energy change
n	number of electrons transferred (in the reaction); [1 mark]
E^θ_{cell}	standard <u>cell</u> potential; [1 mark]
F	Faraday constant; [1 mark]

[Total: 3 marks]

- Careful:** It can be relatively easy to give the incorrect answer for n and E^θ_{cell}

- If the oxidised species or reduced species does not have a concentration then it is replaced by the number 1

4a

a) The completed table is:

Term	
ΔG^θ	Standard free energy change
n	number of electrons transferred (in the reaction); [1 mark]
E^θ_{cell}	standard <u>cell</u> potential; [1 mark]
F	Faraday constant; [1 mark]

[Total: 3 marks]

- **Careful:** It can be relatively easy to give the incorrect answer for n and E^θ_{cell}
 - n is often used to represent the number of moles but in this equation, it is the number of electrons transferred
 - E^θ_{cell} is the standard cell potential not the standard electrode potential

4b

b) The **two** half-cell electrode reactions and their E^θ values are:

- $\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$
AND
– 0.74 (V); [1 mark]
- $\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$
AND
– 0.04 (V); [1 mark]

[Total: 2 marks]

- From the equation given in the question, you should be able to identify that the required half-equations involve:
 - Fe^{3+} and Fe
 - The relevant half-equation is $\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$ with an E^θ value of – 0.04 V
 - Cr and Cr^{3+}

- The relevant half-equation is $\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$ with an E^θ value of -0.74 V

4c

c) To calculate the standard Gibbs free energy change for the electrochemical cell:

- $E^\theta_{\text{cell}} = 0.70 \text{ (V)}$; [1 mark]
- $\Delta G^\theta = -3 \times (+0.70) \times 96\,500$; [1 mark]
- $\Delta G^\theta = -202\,650 \text{ (J mol}^{-1}\text{)}$; [1 mark]
- $\Delta G^\theta = -203 \text{ (kJ mol}^{-1}\text{)}$; [1 mark]

[Total: 4 marks]

- You are given the equation in part (a)
 - $\Delta G^\theta = -n \times E^\theta_{\text{cell}} \times F$
- n is the number of electrons transferred in the reaction
 - Both half-equations transfer 3 electrons, which means that $n = 3$
- E^θ_{cell} is the standard cell potential
 - This is calculated by $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}}$
 - The $\text{Fe}^{3+} / \text{Fe}$ half-equation has the most positive standard electrode potential, so it is the reduction reaction
 - $E^\theta_{\text{cell}} = E^\theta_{\text{reduction}} - E^\theta_{\text{oxidation}} = (-0.04) - (-0.74) = +0.70 \text{ V}$
- F is the Faraday constant, which is given as $96\,500 \text{ C mol}^{-1}$
- Substituting these values into the equation
 - $\Delta G^\theta = -3 \times (+0.70) \times 96\,500$
- Evaluation of the equation
 - $\Delta G^\theta = -202\,650$
 - But this value is in J mol^{-1} , which means that you have to divide by 1000 to get a final answer in kJ mol^{-1}
 - $\Delta G^\theta = -\frac{202\,650}{1000} = -203 \text{ kJ mol}^{-1}$ to 3 significant figures

4d

d) The reaction is:

- Feasible

AND

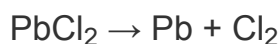
Because the value of ΔG^θ is negative; [1 mark]

[Total: 1 mark]

- **Remember:** A negative value of ΔG^θ means that a reaction is feasible, while a positive value means that the reaction is not feasible

5a

a) The equation for the electrolysis of lead(II) chloride is:



- Correct chemical formulae; [1 mark]

[Total: 1 marks]

- The electrolysis of molten lead(II) chloride will produce lead and chlorine

5b

b)

i) The half-equation for the reduction of the lead(II) ion, Pb^{2+} is:

- $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$; [1 mark]

ii) The number of coulombs required to deposit one mole of lead at the cathode is:

- 193 000 C; [1 mark]

To calculate the charge transferred during the electrolysis of the molten lead(II) bromide:

- $t = 600 \text{ (s)}$; [1 mark]
- Charge transferred, $Q = 990 \text{ C}$; [1 mark]

To calculate the mass of lead deposited at the cathode:

- Moles of lead deposited = 5.1295×10^{-3} ; [1 mark]
- Mass of lead deposited = 1.06 (g); [1 mark]

[Total: 6 marks]

- This question guides you through the steps required to calculate the mass of lead deposited during the electrolysis described in part (a)
- You need to know the number of electrons involved in the electrolysis
 - This comes from the relevant half-equation
 - $\text{Pb}^{2+} + 2\text{e}^- \rightarrow \text{Pb}$
 - There are 2 electrons involved in this electrolysis
- You can use the number of electrons to calculate the number of coulombs required to deposit one mole of lead at the cathode
 - One mole of electrons requires 96 500 coulombs
 - Therefore, $2 \times 96\,500 = 193\,000$ coulombs are required for this electrolysis
- Using the $Q = I \times t$ equation, you can calculate the charge or number of coulombs transferred during the electrolysis
 - However, you have to convert the time from minutes to seconds first
 - $10 \times 60 = 600 \text{ s}$
 - $Q = I \times t = 1.65 \times 600 = 990 \text{ C}$
- You can now use these two values to determine the number of moles of lead deposited
 - Number of moles of lead deposited = $\frac{990}{193000} = 5.1295 \times 10^{-3}$
- Then use $\text{mass} = \text{moles} \times M_r$ to calculate the mass of lead deposited
 - $\text{Mass} = 5.1295 \times 10^{-3} \times 207.2 = 1.06 \text{ g}$ to 3 significant figures

5c

c)

i) The half-equation for the oxidation of the chloride ion, Cl^- to form chlorine, Cl_2 is:

- $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$; [1 mark]

ii) The number of coulombs required to form one mole of chlorine at the anode is:

- 193 000 C; [1 mark]

To calculate the volume of chlorine released at the anode:

- Moles of chlorine released = 5.1295×10^{-3}

AND

Volume of chlorine released = $0.123 \text{ (dm}^3\text{)}$; [1 mark]

[Total: 3 marks]

- Although some of the required steps in this calculation have already been completed in part (b), you are still guided through the steps required to calculate the volume of chlorine released during the electrolysis described in part (a)
- You need to know the number of electrons involved in the electrolysis
 - This comes from the relevant half-equation
 - $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
 - There are 2 electrons involved in this electrolysis
- This results in the same calculation as part (b) for the number of coulombs required to deposit one mole of lead at the cathode
 - 193 000 coulombs are required for this electrolysis
- The charge or number of coulombs transferred during the actual electrolysis is also the same as part (b)
 - $Q = I \times t = 1.65 \times 600 = 990 \text{ C}$
- The number of moles of chlorine released will be the same as the number of moles of lead deposited since both reactions transfer 2 electrons
 - Number of moles of chlorine released = $\frac{990}{193000} = 5.1295 \times 10^{-3}$
- One mole of gas occupies 24.0 dm^3
 - So, 5.1295×10^{-3} of gas will occupy $5.1295 \times 10^{-3} \times 24.0 = 0.123 \text{ dm}^3$ to 3 significant figures